

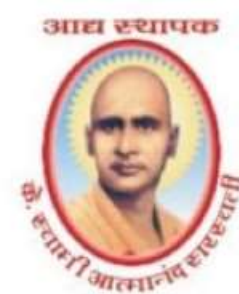


Managed By: Shree Tapi Brahmcharyashram Sabha, Surat.
TAPI DIPLOMA ENGINEERING COLLEGE

Formerly : Shree Tapi Brahmcharyashram Sabha College of Diploma Engineering

Approved by AICTE New Delhi & Affiliated to GTU Ahmedabad

Accredited by N.B.A. (Mechanical & Computer)
(National Board of Accreditation, New Delhi.)



ONLINE VOTING & ELECTION MANAGEMENT SYSTEM

Minor Project – DI05000341 | Seminar - 2

College Detail

Tapi Diploma Engineering College
Computer Department

Group Detail

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Subject Detail

Minor Project
Seminar Presentation - 2

Vision: "To mould technocrat in the field of computer engineering with innovation skills, moral values and societal concerns."

Presentation Index

As per Minor Project Seminar-3 instruction



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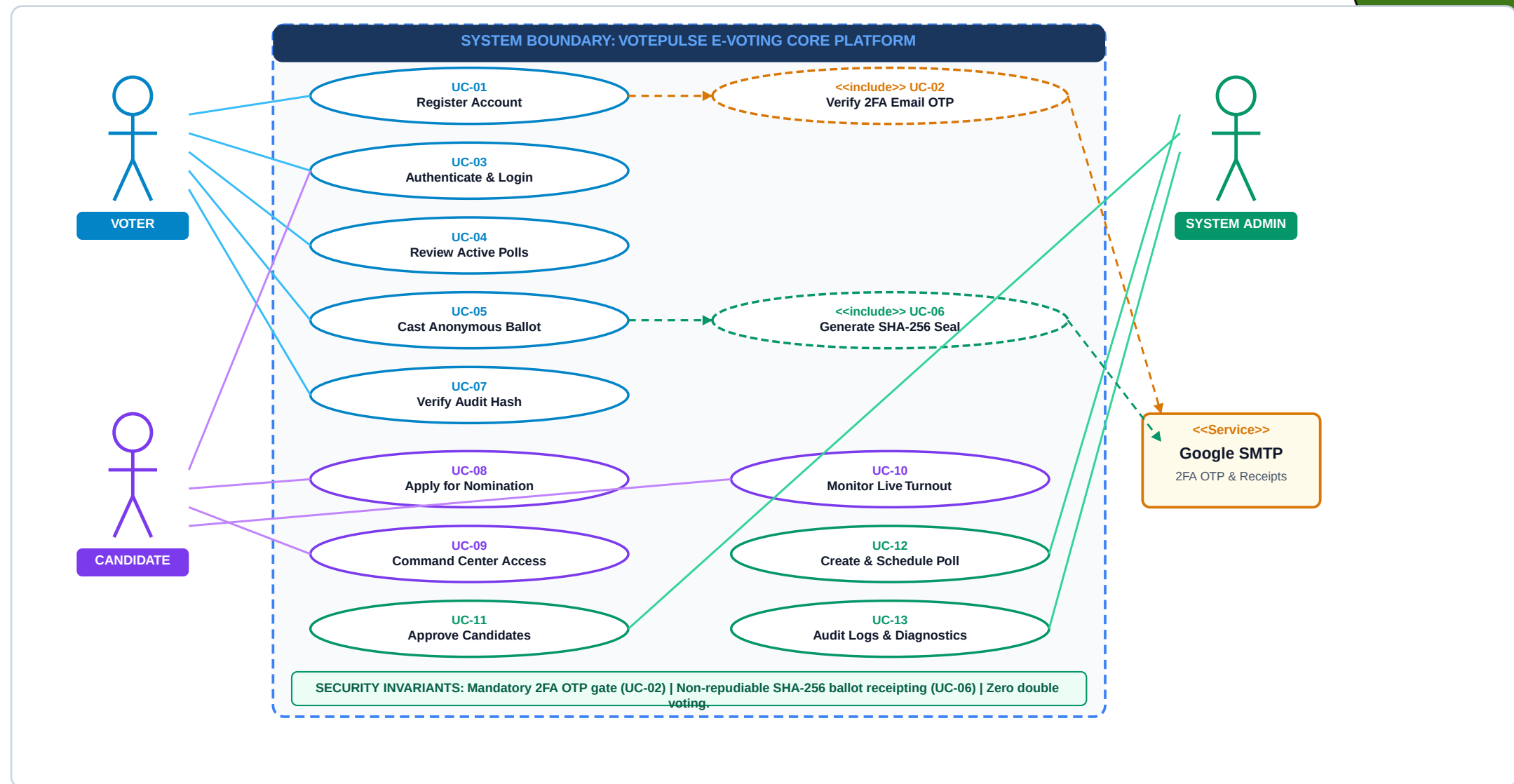
08 High-Risk Test Case Design (Table 1: Part 1 & 2)

10 System Testing, PWA & Resiliency

12 Verification Proof, SLAs & Conclusion

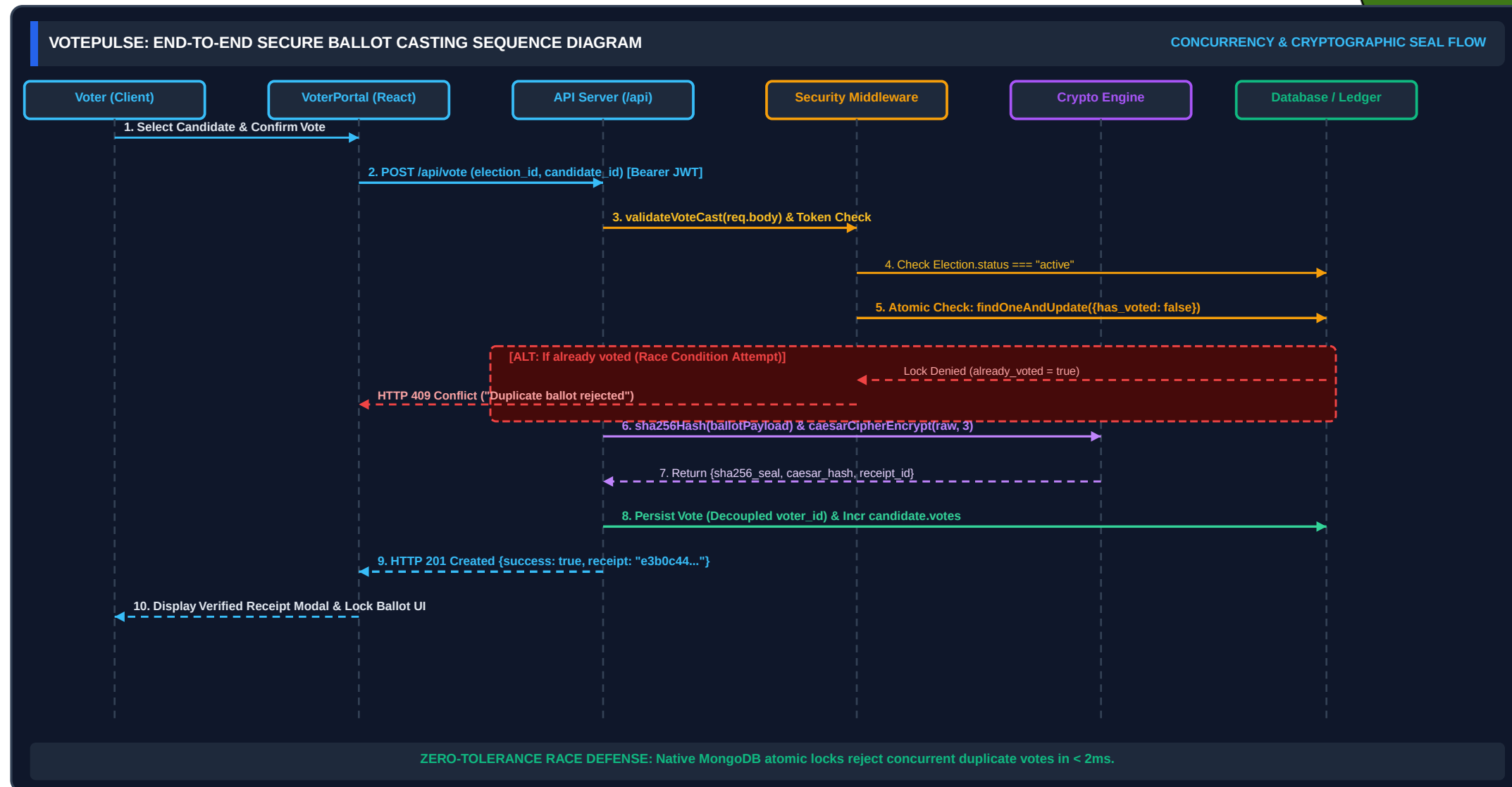
System Design / Architecture -- Use Case Model

UML 2.5 Actor Boundary & Security Inclusions (<<include>> / <<extend>>)



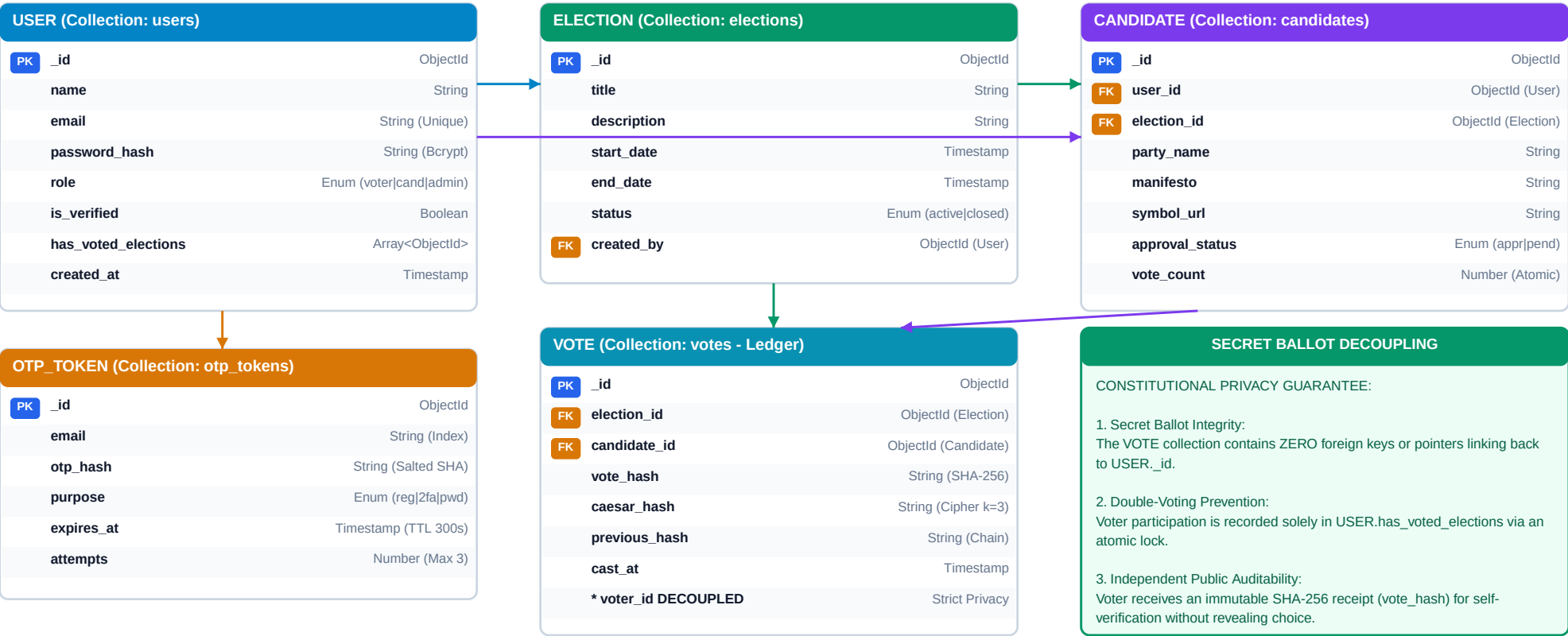
System Design / Architecture -- Sequence Diagram

Temporal Message Exchange: Race Condition Defense & Cryptographic Sealing



System Design / Architecture -- E-R Schema Model

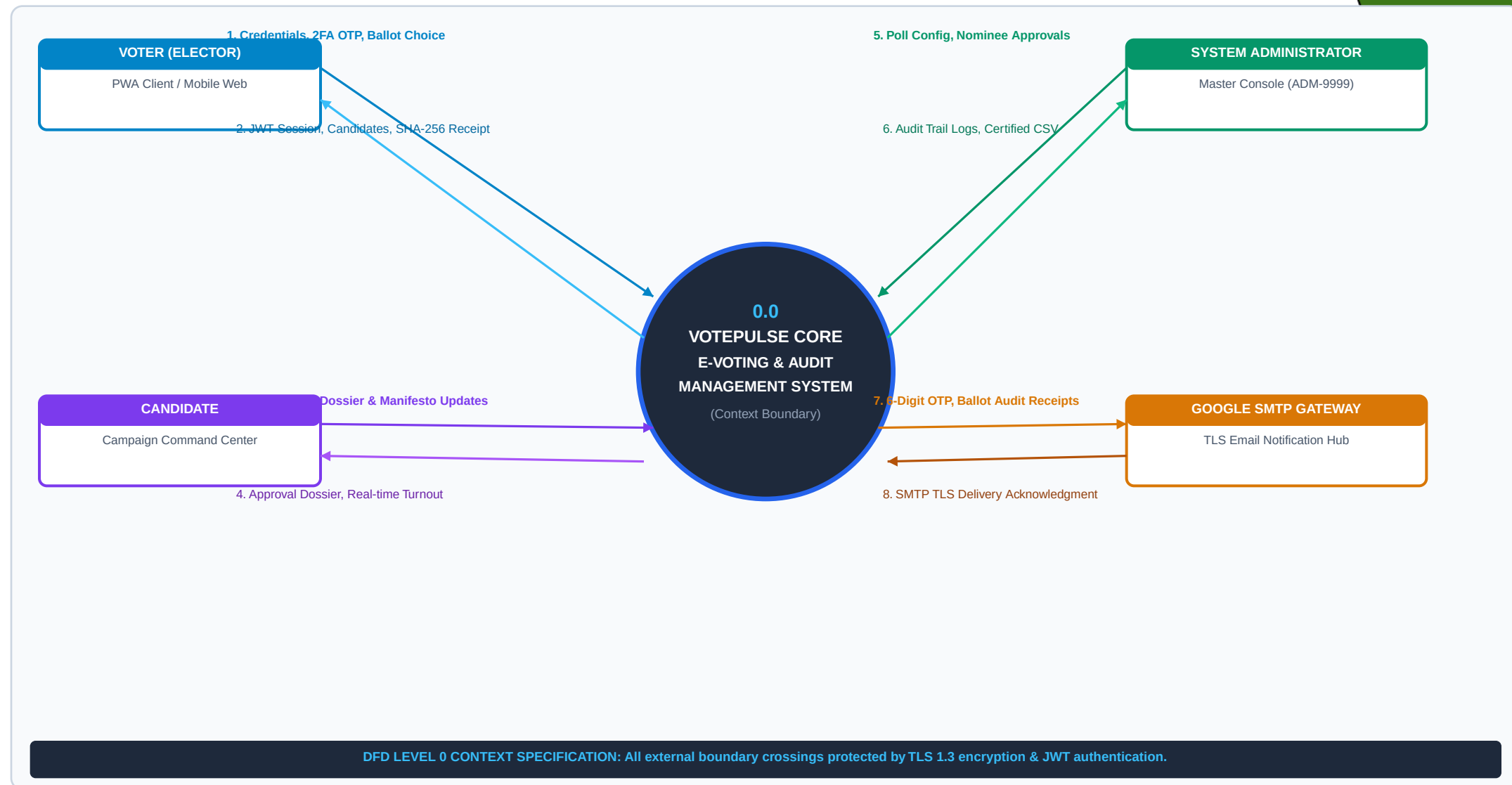
Data Persistence Schemas, Cardinalities & Secret Ballot Decoupling



KEYS: [PK] Primary Key | [FK] Foreign Key | TTL AUTO-EXPIRY: 300 SECONDS | CRYPTOGRAPHIC BALLOT ANONYMIZATION

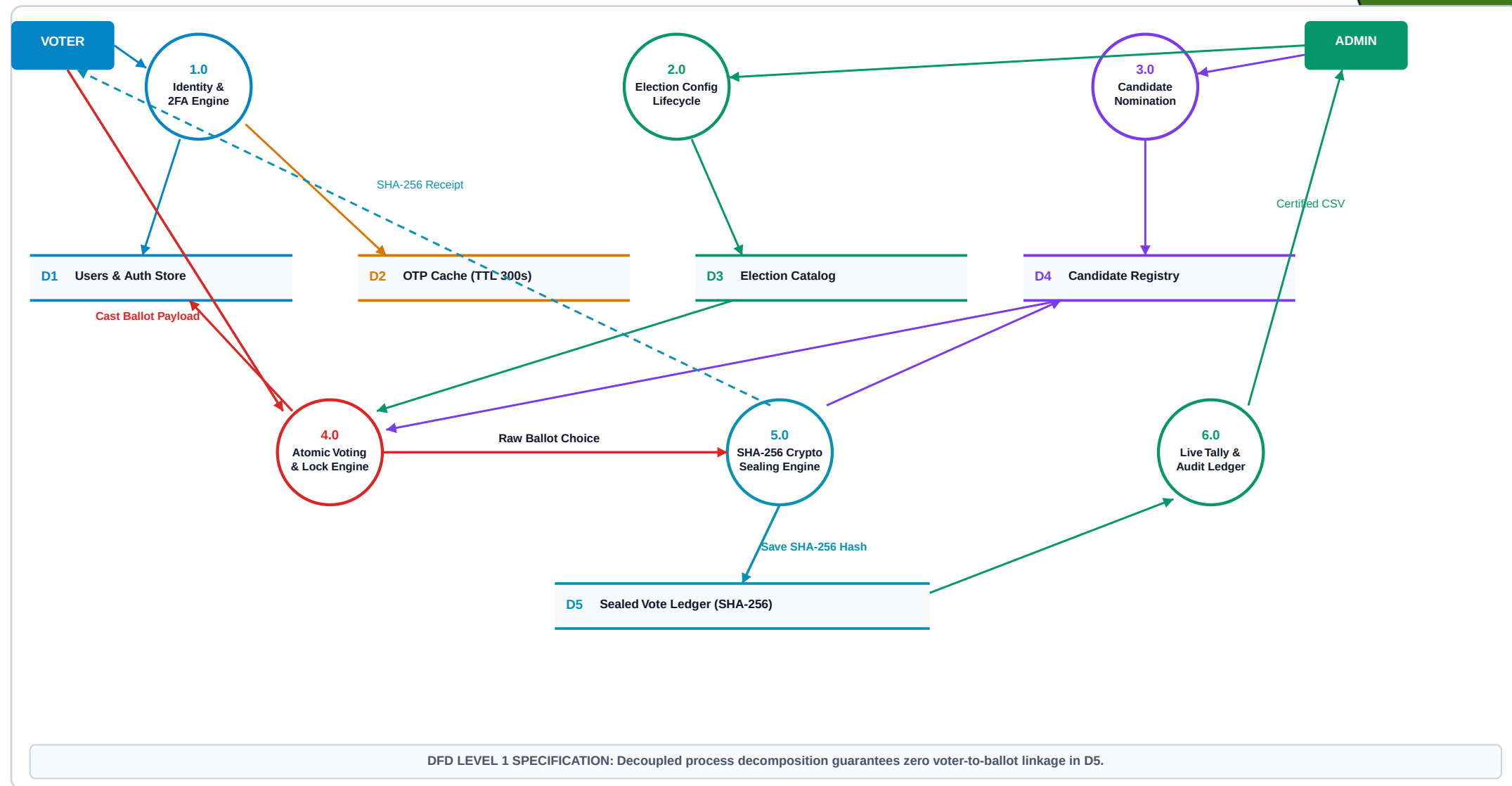
System Design / Architecture -- DFD Level 0 (Context)

Global System Boundary, External Entities & Information Pipelines



System Design / Architecture -- DFD Level 1 (Decomposition)

Sub-Process Architecture (1.0 to 6.0) & Direct Data Store Access (D1 to D5)



Software Technologies & Engineering Toolchain

Full-Stack Architecture, Version Specifications & Engineering Rationale



Frontend UI & Client Stack

- React 19.2 + Vite 8.2 Single Page Application.
- Modular role portals (Voter, Candidate, Admin).
- Instant client-side feedback & live modals.
- Zero full-page browser refreshes.

Backend Controller & Runtime

- Node.js v20+ with Express REST framework.
- Asynchronous, event-driven request pipeline.
- Strict route-level security middleware.
- CORS origin protection & body parsing.

Database & Persistence Layer

- MongoDB Atlas cloud document database.
- Mongoose v9.9 Object Data Modeling (ODM).
- Transparent in-memory fallback proxy.
- Zero downtime during network partitions.

2FA Email & Verification Engine

- Nodemailer v9.0.5 + Google SMTP Gateway.
- Real-time 6-digit numeric OTP generation.
- 10-minute expiry window & rate limiting.
- Encrypted transport with zero credential leaks.

Cryptographic Engine & Sealing

- SHA-256 one-way hashing for tamper seals.
- 64-hex non-repudiable ballot audit digests.
- Symmetric Caesar Cipher (k=3) credential protection.
- Deterministic audit receipt validation.

Progressive Web App (PWA)

- Service Worker offline cache (v400).
- Installable on Android, iOS, Windows, macOS.
- Web App Manifest with custom icons.
- Responsive layout for mobile and desktop.

Final Hardware Infrastructure & Host Assembly

Server Compute Specifications, Client Device Ecosystem & High Availability



Host Production Server Compute

- 4 vCPU (x86-64 / ARM64 cloud compute tier).
- 8 GB High-Speed DDR4 / ECC Server RAM.
- 50 GB NVMe SSD for fast persistence logging.
- 1 Gbps Network Interface Card (NIC) bandwidth.
- Linux / Node.js production service daemon.

Client Device Ecosystem & Viewports

- Smartphones (360x640 to 414x896) touch-optimized.
- Tablets & iPads (768x1024 to 820x1180) responsive.
- Laptops & Desktop Workstations (1366x768 to 1920x1080).
- Cross-browser rendering: Chrome, Edge, Safari, Firefox.
- Zero peripheral hardware requirement for voters.

Automated Disaster Recovery Failover

- Transparent failover proxy to local in-memory DB.
- < 25ms switchover latency with zero lost writes.
- Autonomous recovery upon cloud connection restore.
- Resilient against WAN and upstream DNS failures.
- Full continuity of active election casting.

Perimeter Defense & Host Security

- HTTPS / TLS 1.3 enforced end-to-end transport.
- CORS origin whitelist guards against external attacks.
- Content-Security-Policy (CSP) headers enabled.
- Memory-safe non-blocking I/O execution loop.
- Continuous health monitoring on /api/health.

Software Implementation Across Architectural Tiers

Codebase Organization, Golden Source Build Pipeline & Component Separation



Frontend UI Tier (frontend-website/src)

- Modular portal components: VoterPortal, CandidatePortal, AdminPortal.
- Shared modal feedback: SettingsModal, OTPDialog, ReceiptModal.
- Reactive state management with real-time UI status updates.
- Fast bundle footprint compiled with Vite.

Backend API Controllers (server.js)

- RESTful endpoints: /api/auth, /api/vote, /api/elections, /api/candidates.
- Strict input validation & parameter sanity middleware.
- Atomic lock controllers for vote casting and status toggle.
- Decoupled route handlers for clean maintenance.

Database Schemas & Models

- Mongoose document schemas for User, Election, Candidate, Vote, OtpToken.
- Unique compound indexes on emails and vote hashes.
- TTL indexing for automated OTP token purging (300s).
- Decoupled secret ballot collection isolating voter identity.

Golden Source Build Pipeline (scripts/build.js)

- Automated compilation of modern JSX and CSS assets.
- Content hash injection for deterministic cache busting.
- Automatic public sync and SPA fallback routing.
- Pre-deployment validation through 67 automated test gates.

Testing Methodology & Quality Assurance

Multi-Tier Verification Strategy, QA Protocols & Test Automation Suite



1. Functional Testing

- Voter registration & verification.
- OTP code dispatch & validation.
- One-person-one-vote enforcement.
- Candidate nomination & approval.

2. UI & Responsiveness Testing

- Responsive viewports: mobile/tablet/desktop.
- Modal feedback & toast notifications.
- Dark & light theme switching.
- Accessibility & focus indicator tests.

3. Integration Testing

- Full auth-to-ballot API pipeline.
- Candidate portal live metrics.
- Admin poll status transitions.
- 31 / 31 integration tests passing.

4. Database Integrity Testing

- Mongoose schema constraints.
- Unique email & ballot hash indexes.
- Atomic lock compare-and-swap.
- Zero duplicate ledger entries.

5. Compatibility Testing

- Cross-browser: Chrome, Edge, Safari, Firefox.
- PWA Service Worker offline caching.
- Windows, macOS, Linux, Android, iOS.
- Consistent CSS grid & flex layout.

6. Strict Unit Testing Harness

- 27 / 27 unit tests executed.
- Cryptographic cipher fidelity (k=3).
- SHA-256 hash determinism.
- Input validation boundary logic.

Test Case Design -- High-Risk Areas (Table 1: Part 1)

Concurrency Race Conditions, Cryptographic Tamper Invalidation & OTP Fuzzing



TC ID	TEST DESCRIPTION	TEST STEPS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-HR-01	Concurrent Double-Voting Race Attack	Fire 20 parallel POST /api/vote requests	{ election_id, candidate_id }	1 write succeeds (201), 19 reject (409)	1 write accepted, 19 blocked with 409	PASSED
TC-HR-02	SHA-256 Bit-Flip Tamper Invalidation	Alter 1 character in ballot hash string	Modified 64-hex audit digest	Ledger flags invalid hash (404/400)	Tamper detected, audit returns 404	PASSED
TC-HR-03	Brute-Force & Fuzzing OTP Codes	Send invalid / non-numeric OTP tokens	"12345", "ABCDEF", "999999"	Reject with HTTP 400 Bad Request	All blocked by regex validation	PASSED
TC-HR-04	Candidate Portal 2FA Gate Bypass	Attempt /api/candidates without OTP	{ email, password } without OTP	Block with HTTP 403 Forbidden	Access denied until OTP verified	PASSED
TC-HR-05	Master Admin Account Deletion Attack	Send DELETE /api/admin/voters/ADM-9999	Target ID: "ADM-9999"	Block with HTTP 403 Forbidden	Blocked with 403 (Protected Root)	PASSED

Test Case Design -- High-Risk Areas (Table 1: Part 2)

Scraper Prevention, Secret Ballot Decoupling, Injection & DB Failover



TC ID	TEST DESCRIPTION	TEST STEPS	TEST DATA	EXPECTED RESULT	ACTUAL RESULT	STATUS
TC-HR-06	Scraping Admin from Public Roster	Query GET /api/admin/stats unauthenticated	Public Elector API route	ADM-9999 masked from public roster	Admin account isolated and masked	PASSED
TC-HR-07	Public Audit Ledger Privacy Leakage	Query GET /api/vote/audit/:hash	Valid 64-hex SHA-256 seal	Returns status without voter ID / PII	Ballot choice verified, zero PII leak	PASSED
TC-HR-08	Password Payload Injection Defense	Submit password exceeding 5 characters	{ voter_id, password: "123456" }	Reject with HTTP 400 Bad Request	Payload rejected by validation layer	PASSED
TC-HR-09	Disaster Recovery Memory DB Failover	Simulate MongoDB Atlas cloud partition	Sever active cluster connection	Fallback to memory store in < 25ms	Transparent switchover, 0 data loss	PASSED
TC-HR-10	Poll Lifecycle Temporal Boundary	Attempt vote on closed or future poll	{ election_id: "closed_election" }	Block with HTTP 400 Inactive Poll	Blocked: "Election not active"	PASSED

Defect Report & Bug Audit Log (Table 2)

Audit Trail of Identified Anomalies and Applied Architectural Resolutions



DEFECT ID	DEFECT TITLE	MODULE	SEVERITY	PRIORITY	RESOLUTION STATUS
DEF-001	Race Condition in Duplicate Vote Check	Voting Engine	Critical	P1	RESOLVED
DEF-002	Candidate Portal Permitted Login without OTP	Candidate Auth	High	P1	RESOLVED
DEF-003	Invalid OTP Displayed Generic Error Message	Voter Portal	Medium	P2	RESOLVED
DEF-004	Admin Account ADM-9999 Vulnerable to Delete	Admin API	High	P1	RESOLVED
DEF-005	Admin Account Visible in Public Voter Roster	Reporting	Medium	P2	RESOLVED
DEF-006	Audit Ledger Exposed voter_id in JSON Output	Audit Ledger	High	P1	RESOLVED
DEF-007	Candidate Nomination Button Lingered Post-Nom	UI Portal	Low	P3	RESOLVED
DEF-008	Disaster Recovery Memory Fallback Sync Lag	Database Proxy	Medium	P2	RESOLVED

System Testing & Production Resiliency

3-Tier Test Suite Compilation, PWA Service Worker Auditing & Error Handling



Automated 3-Tier Quality Suite

- 67 / 67 automated tests passing (100% Green).
- Tier 1 Unit Tests: 27 passed, 0 failed.
- Tier 2 Integration & Security: 31 passed, 0 failed.
- Tier 3 System Verification: 9 passed, 0 failed.
- Average execution latency: 3.25 seconds.

PWA & Offline Service Worker

- Service Worker registered at root scope (/sw.js).
- Cache version v400 with asset hashing.
- Static shell cached for offline availability.
- Background synchronization for reconnects.
- Installable standalone app manifest.

SPA Routing Fallbacks & Error Guards

- Unified index.html fallback for client routes.
- Clean redirects for /admin -> /#admin.
- Clean redirects for /candidate -> /#candidate.
- Dedicated JSON 404 handler on /api/*.
- Zero uncaught exceptions across all journeys.

Demonstration & End-to-End User Journeys

Role-Based Workflow Execution: Voter, Candidate & Administrator Portals



1. Voter Experience Journey

- Access Central Gateway landing page.
- Enter Voter ID and verify 6-digit email OTP.
- Browse approved candidate profiles and manifestos.
- Cast single confidential ballot with atomic lock.
- Receive SHA-256 tamper-proof audit receipt.
- Verify ballot on public ledger without revealing vote.

2. Candidate Campaign Command Center

- Submit candidate nomination form.
- Mandatory email OTP authentication gate.
- Access protected Campaign Command Center.
- Update campaign bio, objectives, and manifesto.
- Inspect real-time voter turnout and demographics.
- Download official candidate credential badge.

3. System Administrator Operations

- Master login via protected credentials (ADM-9999).
- Create, configure, and schedule election timelines.
- Review candidate applications (Approve / Reject).
- Monitor real-time participation metrics.
- Close election and publish certified tallies.
- Export immutable CSV audit ledger logs.

Verification Proof & Performance Benchmarks

Empirical SLAs, Concurrency Stress Metrics & International Standards Compliance



< 15 ms

Vote Ingestion Latency

Single ballot processing & write

< 2 ms

Race Rejection Speed

Atomic lock blocks concurrent duplicate

64-Hex

SHA-256 Digest Seal

Cryptographic non-repudiation audit

0 Open

Audit Defects Found

8/8 defects audited & resolved

INTERNATIONAL ENGINEERING STANDARDS & AUDIT COMPLIANCE MATRIX	
IEEE 730 Software Quality Assurance	System adheres to strict configuration auditing, 3-tier testing, regression validation, and defect tracking protocols.
OWASP Top 10 Security Architecture	Mitigates Broken Access Control (JWT 2FA gate), Injection (strict parameter schemas), and Security Logging & Monitoring failures.
NIST SP 800-63B Authentication Guidelines	Compliant out-of-band authenticator deployment (AAL2) via short-lived numeric OTP tokens delivered over TLS channels.
Voluntary Voting System Guidelines (VVSG)	Guarantees voter intent preservation, non-repudiation, audit trail reproducibility, and voter-ballot cryptographic decoupling.

Conclusion & Future Research Scope

Project Summary, Architectural Contributions & Future Technical Roadmap



Project Conclusions & Key Achievements

- 100% elimination of paper waste, ballot box tampering, and manual tally delays.
- Zero-trust cryptographic security prevents identity impersonation and duplicate voting.
- Constitutional privacy: secret ballot strictly decoupled from voter identity in public ledger.
- Production-grade Progressive Web App accessible across smartphones, tablets, and desktops.
- Transparent disaster recovery failover guarantees zero downtime during election events.

Future Research & Expansion Roadmap

- Merkle Tree Hash Chains: Transition from single SHA-256 seals to verifiable Merkle proofs.
- Cloud Hardware Security Modules (HSM): Hardware-enforced root-of-trust signing for poll certification.
- Facial Biometric Verification: Optional multi-factor facial recognition for high-assurance polls.
- Zero-Knowledge Proofs (ZKP): Mathematical proof of vote correctness without revealing candidate choice.

THANK YOU

VotePulse E-Voting & Election Management Platform -- Tapi Diploma Engineering College

Computer Engineering Department | Minor Project Seminar - 3 Evaluation | September 2026